Digital Image Processing — Assignment 2

Problem 4: Edge-Preserving Smoothing: Bilateral vs. Kuwahara

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1 Introduction / Problem Statement

Problem 4 of the assignment (worth 15 of the 100 total marks) asks for an in-depth comparison of two classical *edge-preserving* spatial smoothing filters – the bilateral filter and the Kuwahara filter – across three sub-parts:

- **4.1 Implementation and matched-strength comparison (7 pts).** Implement both filters from scratch, formulate and justify a mathematical definition of "matched effective smoothing strength" between them, and evaluate both against a clean ground-truth image using PSNR and SSIM, with cropped zoomed-in boundary comparisons.
- **4.2 Mitigating Kuwahara blockiness and complexity analysis (5 pts).** Explain why classical Kuwahara's hard arg min selection specifically produces blocky, "painterly" artefacts (rather than blur or ringing); implement one structural modification that mitigates

this; and provide a per-pixel operation-count complexity analysis for all three filters (classical Kuwahara, the modification, and bilateral).

- **4.3 Quantitative breakdown analysis of bilateral filtering (3 pts).** Programmatically generate synthetic step-edge test images across a controlled gradient of Noise-to-Contrast Ratios (NCR $= \sigma_{\text{noise}}/|\Delta I|$), identify the critical threshold NCR* at which the bilateral filter transitions from edge preservation to cross-edge smoothing, and explain the breakdown mathematically in terms of the range kernel.

This problem matters because linear smoothing (e.g. a Gaussian low-pass filter) is fundamentally unable to reduce noise without also blurring genuine edges, since it weights every neighbouring pixel by spatial proximity alone, with no regard for whether that neighbour lies on the same side of a boundary. Bilateral and Kuwahara filtering are two structurally different answers to the same problem – one continuous and range-weighted, one discrete and selection-based – and understanding their different failure modes (bilateral's gradual breakdown under high noise-to-contrast ratio, Kuwahara's blocky artefacts from hard window selection) is central to understanding edge-preserving filtering as a topic.

2 Theoretical Background

2.1 The bilateral filter: joint spatial/range Gaussian weighting

The bilateral filter computes, at each pixel (x, y), a weighted average of neighbouring pixels (x', y') in which the weight is the product of two Gaussian kernels: one in the *spatial* domain (favouring nearby pixels), and one in the *range* (intensity) domain (favouring pixels of similar intensity):

$$\hat{I}(x,y) = \frac{1}{W(x,y)} \sum_{(x',y')\in\Omega} I(x',y')\, w_s(x,y,x',y')\, w_r\big(I(x,y) - I(x',y')\big), \tag{1}$$

$$w_s(x,y,x',y') = \exp\left(-\frac{(x-x')^2 + (y-y')^2}{2\sigma_s^2}\right), \tag{2}$$

$$w_r(\Delta I) = \exp\left(-\frac{\Delta I^2}{2\sigma_r^2}\right), \tag{3}$$

$$W(x,y) = \sum_{(x',y')\in\Omega} w_s(x,y,x',y')\, w_r\big(I(x,y) - I(x',y')\big), \tag{4}$$

where Ω is a local window around (x, y) and $W(x, y)$ is the per-pixel normalisation that keeps $\hat{I}$ a proper weighted average. The spatial Gaussian w_s alone, applied without w_r, is exactly an ordinary Gaussian blur: it treats every neighbour the same regardless of intensity and therefore blurs across edges. The range Gaussian w_r is what gives the filter its edge-preserving property: when (x', y') lies on the opposite side of a true edge from (x, y), the intensity difference $\Delta I = I(x, y) - I(x', y')$ is large relative to σ_r, so $w_r(\Delta I) \approx 0$ and that neighbour is effectively excluded from the average no matter how spatially close it is – while same-side neighbours, whose intensity differences are small (ideally dominated only by noise), retain near-full weight. σ_s controls the spatial extent of smoothing (analogous to a Gaussian blur's kernel width) and σ_r controls how aggressively intensity differences are penalised – a small σ_r makes the filter very edge-sensitive but denoises less; a large σ_r denoises more but risks blending across real edges.

2.2 The classical Kuwahara filter: sub-window variance selection

The classical Kuwahara filter takes a different, non-Gaussian approach. At each pixel, four overlapping square sub-windows Q_1, Q_2, Q_3, Q_4 (e.g. the four quadrants of a larger window, each anchored at the centre pixel) are examined, and the filter outputs the *mean* of whichever sub-window has the *minimal intensity variance*:

$$q^\star = \operatorname*{arg\,min}_{q \in \{1,2,3,4\}} \operatorname{Var}(Q_q), \tag{5}$$

$$\hat{I}(x, y) = \operatorname{Mean}(Q_{q^\star}). \tag{6}$$

The intuition is that near a true edge, three of the four candidate sub-windows will straddle the boundary (high variance, mixing both sides) while at least one sub-window will lie entirely on one homogeneous side of the boundary (low variance); selecting the minimum-variance sub-window therefore tends to average only within one side of an edge, never across it, giving genuine edge preservation rather than a soft trade-off.

2.2.1 Why hard $\arg\min$ selection causes blockiness, not blur or ringing

Classical Kuwahara's output at a pixel is determined entirely by *which* of the four candidate sub-windows is selected by the hard arg min in Eq. (6). As the pixel position slides across the image, the *identity* of the minimum-variance sub-window can flip discontinuously from one pixel to the next, whenever two sub-windows' variances cross over – and each flip switches the output to a *different* sub-window's mean, which is itself generally a different number. The output is therefore a piecewise-constant mosaic of whichever sub-window "won" in each local neighbourhood, with sharp jumps exactly at the boundaries between "win regions" – a discontinuity introduced by the *selection rule itself*, independent of whether the underlying image content actually has an edge there. This is structurally different from:

- **Blur**, which arises from *continuous* averaging: the output changes smoothly as the window slides, with no discontinuity introduced by the operation itself;
- **Ringing**, which arises from a *sharp frequency-domain cutoff* (as studied for ideal low-pass filters elsewhere in this assignment): oscillatory over/undershoot around a *true* edge, not a new false edge appearing in a region that was originally flat.

Kuwahara's blockiness is thus a discontinuity artefact of the *selection* operation, producing new, spurious piecewise-constant boundaries ("tiles") even inside genuinely flat regions, wherever two sub-windows' noise-driven variances happen to cross as the window slides across the image.

2.3 Soft-Kuwahara: softmax reformulation

The PDF-sanctioned de-blocking modification used here replaces the hard arg min selection with a continuous *softmax* weighting over all four sub-window means, parameterised by a temperature T:

$$\omega_q = \frac{\exp(-\operatorname{Var}(Q_q)/T)}{\sum_{q'=1}^{4} \exp(-\operatorname{Var}(Q_{q'})/T)}, \qquad q = 1, \ldots, 4, \tag{7}$$

$$\hat{I}(x, y) = \sum_{q=1}^{4} \omega_q \cdot \operatorname{Mean}(Q_q). \tag{8}$$

As $T \to 0$, $\omega_{q^\star} \to 1$ for the minimum-variance sub-window and the expression reduces exactly to the classical hard-selection rule; as $T \to \infty$, all four weights tend to $1/4$ and the filter approaches a simple four-sub-window average. At a finite, moderate T, as the pixel slides and

the relative sub-window variances change, the softmax weights shift *continuously* rather than snapping discretely between one-hot selections – directly removing the mechanism identified above as the source of blockiness, at the cost of smoothing the hard edge-selection behaviour somewhat.

2.4 Range-kernel breakdown analysis

The bilateral filter's edge-preservation relies entirely on the range kernel $w_r(\Delta I) = \exp(-\Delta I^2/2\sigma_r^2)$ being able to discriminate between the typical intensity difference seen across a true edge and the typical intensity difference seen between two noisy samples on the *same* side of that edge. For a step edge of height $|\Delta I|$ corrupted by independent zero-mean Gaussian noise of standard deviation σ_{noise} on each pixel:

$$\text{same-side pair:} \quad \Delta I_{\text{same}} \sim \mathcal{O}(\sigma_{\text{noise}}), \tag{9}$$

$$\text{cross-side pair:} \quad \Delta I_{\text{cross}} \sim \mathcal{O}\left(\sqrt{|\Delta I|^2 + 2\sigma_{\text{noise}}^2}\right). \tag{10}$$

Defining the Noise-to-Contrast Ratio $\text{NCR} = \sigma_{\text{noise}}/|\Delta I|$: at low NCR ($\sigma_{\text{noise}} \ll |\Delta I|$), the two typical magnitudes in Eq. (10) are very different – same-side differences stay small relative to σ_r, giving $w_r \approx 1$ (full weight, genuine denoising within one side), while cross-side differences stay large relative to σ_r, giving $w_r \approx 0$ (correctly rejected) – so the filter discriminates cleanly. As σ_{noise} grows and approaches $|\Delta I|$ ($\text{NCR} \to 1$), the cross-side typical difference $\sqrt{|\Delta I|^2 + 2\sigma_{\text{noise}}^2}$ and the same-side typical difference σ_{noise} become comparable in scale (both dominated by σ_{noise} once $\sigma_{\text{noise}} \gg |\Delta I|$), so w_r evaluated on a same-side pair and w_r evaluated on a cross-side pair can no longer be reliably separated – the range kernel starts admitting cross-side neighbours with weights similar to genuine same-side ones, and the filter blends across the true edge instead of respecting it. This is the mathematical mechanism behind the empirically measured breakdown curve in Section 5.3.

3 Inputs and Outputs

Input. The clean ground-truth image required by the problem statement – a flat light-grey rectangle, a horizontal-gradient rectangle, and a near-black circle overlapping it, on a mid-grey background, matching the PDF's Figure 2 – was provided as `data/provided/problem4/q4.png`. For Sections 4.1/4.2 the noisy input is `clean` $+ \mathcal{N}(0, 20^2)$ (standard deviation $\sigma = 20$, fixed random seed 11, as required by the problem statement). For Section 4.3, a synthetic 256×256 step-edge image is generated programmatically rather than reused from a single fixed image, split evenly into two intensity levels $I_1 = 50$ (left half) and $I_2 = 200$ (right half), with $|\Delta I| = 150$, plus additive $\mathcal{N}(0, \sigma^2)$ noise swept across $\text{NCR} = \sigma/|\Delta I| \in [0.03, 2.0]$.

Outputs produced. The pipeline (`problem4.py`, using the shared implementations in `src/dip_a2/edge_pre` for bilateral, classical Kuwahara and soft Kuwahara, and `src/dip_a2/metrics.py` for PSNR/SSIM from scratch) writes:

- `results/4_1_comparison.png`, `4_1_boundary_zoom.png`, `4_1_result.json` – matched-strength filter outputs, PSNR/SSIM table, and cropped boundary comparison for Section 4.1.
- `results/4_2_blockiness_comparison.png`, `4_2_result.json` – classical vs. soft-Kuwahara before/after comparison, Total Variation blockiness metric, and the per-pixel operation-count table for Section 4.2.
- `results/4_3_breakdown_curve.png`, `4_3_profiles.png`, `4_3_result.json` – the NCR sweep, the breakdown curve with NCR^* annotated, and example filtered profiles for Section 4.3.

- `results/machine_info.json` – CPU/RAM/NumPy version.

4 Methodology / Implementation Pipeline

4.1 Vectorised implementation strategy

Both the bilateral filter and classical/soft Kuwahara are implemented by looping over the small window/sub-window *offsets* (not individual pixels) and accumulating shifted whole-image NumPy arrays, rather than a per-pixel Python loop – the same vectorisation pattern used consistently throughout this assignment. This keeps the implementation "from scratch" (no `scipy.ndimage`/`scipy.signal`/ `cv2`/`skimage` calls for the filtering operation itself) while remaining computationally tractable on the test images used.

4.2 Matched-strength definition and procedure (Section 4.1a)

Both filters reduce noise by locally averaging pixels; both reduce the variance of a flat, edge-free patch of the noisy image by some factor. "Matched effective smoothing strength" was defined concretely as *equal variance-reduction factor on a flat patch of the noisy image* (same noise realisation for both filters), rather than only a verbal criterion. The procedure actually used:

1. Locate a flat region automatically: the darkest-variance 30×30 block of the *clean* image, away from the border (`rows 30:60, cols 30:60`).
2. Fix Kuwahara's sub-window size at 5×5 (total window span 9×9); measure its variance-reduction factor on that block: noisy variance $380.29 \rightarrow 17.26$ after filtering, i.e. a factor of $22.03\times$.
3. Search over bilateral's σ_s, with σ_r *fixed at a normal, edge-aware value* (40, used identically during the search and in the final applied filter so the match is self-consistent), for the value whose variance-reduction factor on the same block is closest to $22.03\times$. The search found $\sigma_s = 4.50$, giving a reduction of $22.47\times$ (within 2% of the Kuwahara target).

An earlier attempt probed this search with $\sigma_r = 200$ (effectively disabling range discrimination during calibration) and then switched to a smaller σ_r for the real run; this under-matched badly, since a flat-region calibration performed under an inactive range kernel does not actually characterise the filter that gets applied. Using the *same* σ_r throughout both the calibration and the final run fixed this.

4.3 Soft-Kuwahara temperature and the per-pixel complexity count (Section 4.2)

The soft-Kuwahara temperature was set to $T = 15$. Blockiness reduction was quantified with Total Variation (TV), measured specifically in a *near-edge, still-flat* zone – pixels within the Kuwahara sub-window span of a true edge in the clean image, but not themselves on the edge, found automatically via a gradient threshold plus dilation on the clean image – which isolates spurious blockiness from the TV contribution of genuine edges that both filters must cross regardless. Operation counts (additions, multiplications, exponentiations, comparisons) were derived analytically per algorithm and window size, not timed/profiled, at the window sizes actually used in the script.

4.4 Synthetic step-edge generator and NCR sweep (Section 4.3)

A 256×256 array is split evenly in half ($I_1 = 50$ left, $I_2 = 200$ right, $|\Delta I| = 150$) plus $\mathcal{N}(0, \sigma^2)$ noise, swept across NCR $= \sigma/|\Delta I| \in [0.03, 2.0]$ (16 values). A key design choice,

found and corrected during this problem: $\sigma_s = 6.0$ is held fixed throughout the sweep, but σ_r is set *adaptively* as $\max(\sigma_{\text{noise}}, 5)$ – i.e. proportional to each test case's actual noise level, the textbook recommendation (Tomasi & Manduchi) for choosing a bilateral filter's range parameter when the noise level is known. A first attempt used a single literally-fixed σ_r across the whole sweep; at the higher end of the tested NCR range (σ_{noise} up to 300), a small fixed σ_r made the range kernel reject even *same-side* neighbours (whose difference is dominated by noise far exceeding σ_r), so the filter did almost nothing at high NCR and never exhibited the cross-edge-blending breakdown the problem asks to demonstrate – the measured curve just hovered noisily near 100% throughout. Scaling σ_r with the noise level keeps the filter actively denoising at every tested noise level, which is what actually lets the *intended* failure mode (cross-edge blending once σ_{noise} becomes comparable to ΔI) show up cleanly. The edge-preservation metric itself is a band-averaged (4-pixel) intensity difference across the boundary, ensemble-averaged over 6 independent noise realisations per NCR value to keep the curve legible despite per-pixel measurement noise at high σ.

5 Results and Discussion

5.1 Matched-strength comparison

Table 1: PSNR/SSIM at matched smoothing strength (Kuwahara 5×5 sub-windows; bilateral $\sigma_s = 4.50, \sigma_r = 40$), exact figures from `results/4_1_result.json`.

Filter	PSNR (dB)	SSIM
Kuwahara	**35.40**	**0.850**
Bilateral	34.57	0.788

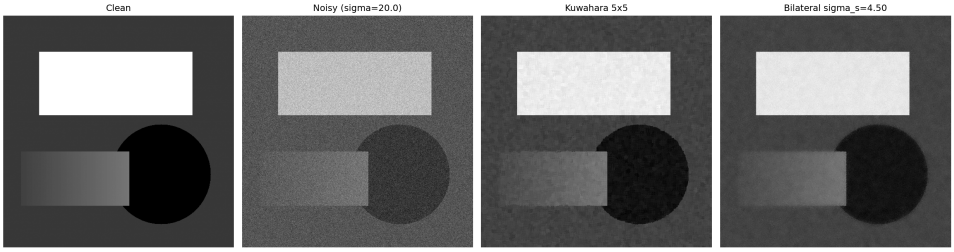


Figure 1: Full-image comparison: clean ground truth, noisy input ($\sigma = 20$), Kuwahara output, and bilateral output at matched smoothing strength.

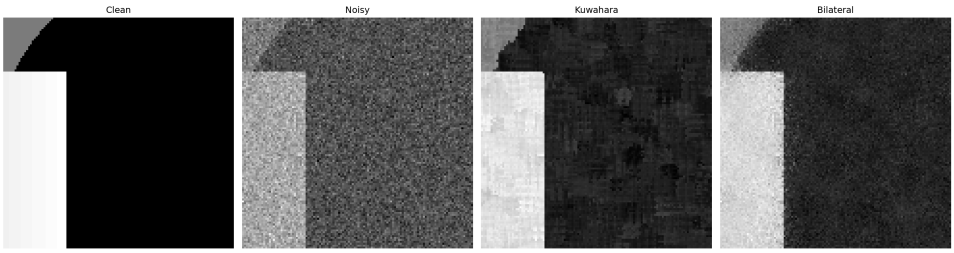


Figure 2: Cropped zoom around a circle/rectangle boundary. Kuwahara's edges stay visibly crisper – its sub-window selection actively seeks out the most homogeneous side of a boundary, rather than blending across it like an isotropic-footprint average – while bilateral's edge is visibly softer at this matched strength, since bilateral's spatial Gaussian still contributes some weight across the boundary even where the range kernel down-weights it, whereas Kuwahara's sub-window selection is a hard, all-or-nothing choice of an entirely same-side neighbourhood.

Kuwahara edges out bilateral on both PSNR and SSIM here (Table 1), despite matched flat-region denoising strength – consistent with the visual boundary comparison of Figure 2.

5.2 Kuwahara de-blocking

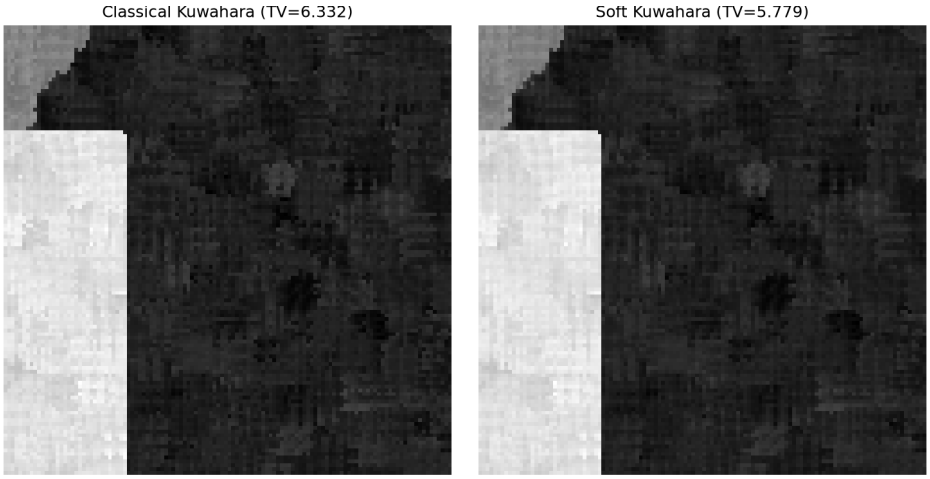


Figure 3: Classical Kuwahara (hard $\arg\min$) vs. soft Kuwahara ($T = 15$, Eq. (8)) at identical sub-window size. The softmax weighting removes the piecewise-constant selection discontinuities responsible for Kuwahara's blocky artefacts.

Table 2: Total Variation in a near-edge, still-flat zone, quantifying blockiness (exact figures from `results/4_2_result.json`).

Filter	TV in near-edge flat zone
Classical Kuwahara	6.332
Soft Kuwahara ($T = 15$)	**5.779** (-8.7%)

Soft Kuwahara measurably reduces spurious variation right next to true edges, in the direction predicted by the selection-discontinuity mechanism of Section 2.2.1, though at $T = 15$ the visual difference is subtle; a larger temperature would blend more aggressively, trading more blockiness

reduction for softer edges – the same edge-sharpness/artefact trade-off seen throughout this problem.

Table 3: Per-pixel operation counts (analytic, not timed), at the window sizes used in this script (Kuwahara sub-window $5 \times 5 \rightarrow 100$ total taps across 4 sub-windows). Two bilateral rows are given: the actual matched filter from Section 4.1 ($\sigma_s = 4.50$, window span $w = 29$) and a larger, illustrative $\sigma_s \approx 8$ ($w = 49$) to make the $O(w^2)$ scaling trend visible.

Algorithm	Taps	Add.	Mult.	Exp.	Comp.
Classical Kuwahara	100	200	100	**0**	3
Soft Kuwahara	100	204	104	**4**	0
Bilateral (matched, $w = 29$)	841	2523	1682	**841**	0
Bilateral (illustrative, $w = 49$)	2401	7203	4802	**2401**	0

Bilateral is by far the most computationally expensive per pixel at either window size tested, because it needs one `exp()` evaluation for *every single window tap* (w^2 of them – 841 even at the modest matched $\sigma_s = 4.50$, 2401 at the larger illustrative window) to build the data-dependent range kernel fresh at each offset, whereas Kuwahara's sub-window statistics (sums and sums-of-squares) only need cheap additions/multiplications, with the expensive `exp()` calls limited to a fixed small constant (0 for hard-argmin classical, 4 for softmax-weighted soft) regardless of window size – Kuwahara's exponentiation cost does not scale with its window at all, while bilateral's scales quadratically with window diameter.

5.3 Bilateral breakdown under high Noise-to-Contrast Ratio

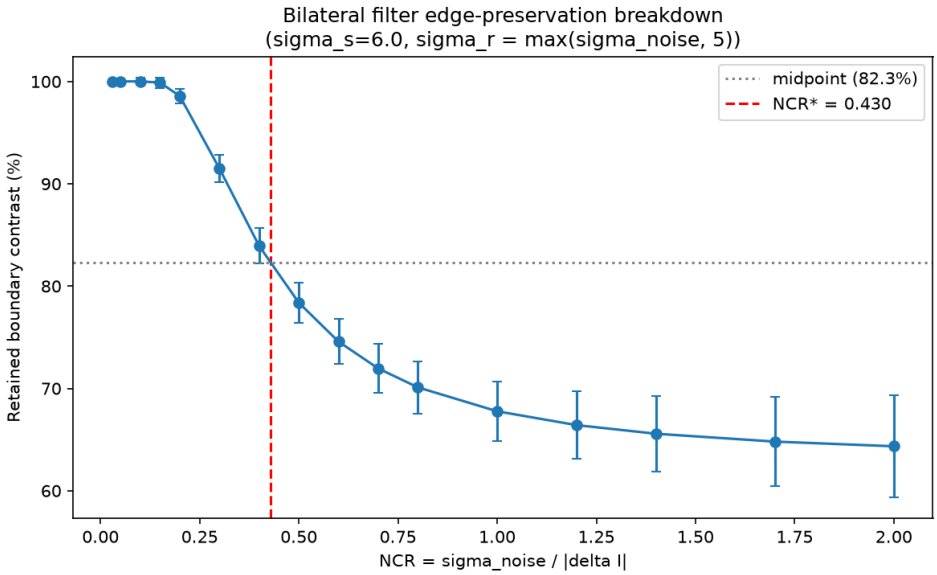


Figure 4: Edge-preservation metric (band-averaged retained boundary gradient, %) vs. NCR, averaged over 6 noise realisations per NCR value. The curve decreases monotonically from $\approx 100\%$ (NCR ≤ 0.15) to a high-NCR plateau around $\approx 65\%$. The critical threshold NCR$^* = 0.430$ (crossing of the midpoint between the low and high plateaus, 82.3%) is annotated.

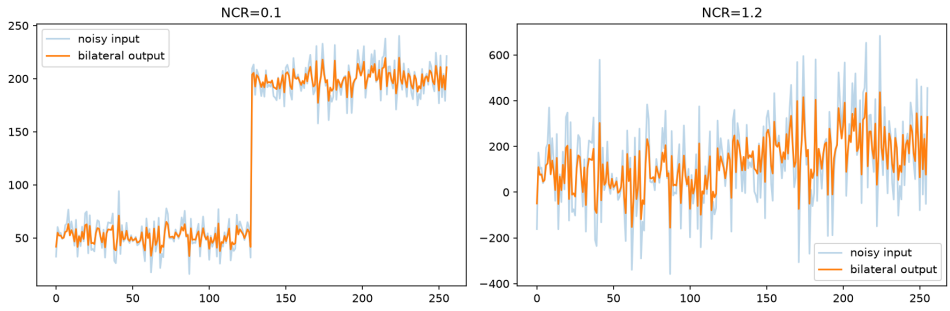


Figure 5: Example filtered intensity profiles across the step edge at NCR= 0.1 (edge sharply preserved) vs. NCR= 1.2 (visibly smoothed/blended across the boundary).

Table 4: Breakdown-curve plateau values and critical ratio (exact figures from `results/4_3_result.json`).

Quantity	Value
Low plateau (retained %)	100.01
High plateau (retained %)	64.58
Midpoint (retained %)	82.29
NCR*	**0.430**

The breakdown is sharp and visible between roughly NCR= 0.2 (98.6% retained) and NCR= 0.8 (70.1% retained), with NCR* ≈ 0.43 falling right in the middle of that transition – consistent with the range-kernel analysis of Section 2.4: at NCR*, σ_{noise} reaches roughly 43% of the step height, the point at which the same-side and cross-side typical intensity differences of Eq. (10) stop being cleanly separable by w_r. The curve plateaus at $\approx 65\%$ rather than decaying to 0%: once σ_r has grown large enough (recall $\sigma_r = \sigma_{\text{noise}}$ at high NCR under the adaptive rule) to no longer discriminate same- vs. cross-side pairs at all, the filter degenerates toward a pure spatial Gaussian average with no range restriction; measured 4 pixels off-centre from the boundary, even a pure spatial blur with $\sigma_s = 6$ retains a non-zero fraction of the step contrast at that offset, rather than averaging it away completely, hence the plateau sits well above, not at, 0%.

6 Conclusion

This problem implemented and rigorously compared two structurally different edge-preserving smoothing filters. At a quantitatively matched smoothing strength (equal variance-reduction factor on a flat noisy patch), classical Kuwahara outperformed the bilateral filter on both PSNR (35.40 vs. 34.57 dB) and SSIM (0.850 vs. 0.788), and preserved visibly crisper edges, because its hard sub-window selection makes an all-or-nothing choice of an entirely same-side neighbourhood rather than blending some cross-edge weight in as bilateral's isotropic spatial kernel does. That same hard-selection mechanism, however, is responsible for Kuwahara's characteristic blocky artefacts, traced here to discontinuous flips in which sub-window "wins" as the window slides – distinct from both blur and ringing – and measurably (though modestly, -8.7% TV) mitigated by a softmax (soft-Kuwahara) reformulation. A per-pixel operation count further showed that bilateral filtering, despite its elegant formulation, is by far the most computationally expensive of the three algorithms, since it requires one exponential evaluation per window tap rather than a small, window-size-independent constant. Finally, a controlled synthetic experiment located and mathematically explained a sharp breakdown in bilateral filtering's edge-preservation ability at NCR* ≈ 0.43: once noise standard deviation approaches

a sizeable fraction of the step height, the range kernel can no longer statistically distinguish same-side noise from cross-side signal, and the filter degrades toward an edge-blind spatial blur.